Optimal Reconfiguration of Systems in Aeronautics

Lecture 11: "Approximate Model Matching"

Bogdan D. Ciubotaru

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Outline

1. Pseudo-Inverse vs. Modified Pseudo-Inverse Methods
2. B747 Application Example

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1. Pseudo-Inverse vs. Modified Pseudo-Inverse Methods
   - Preliminaries on Matrix Inverses and Subspaces
   - Classical Pseudo-Inverse Method
   - Modified Pseudo-Inverse Method
2. B747 Application Example

Preliminaries I

This introduction provides some results on generalized inverses of matrices. It is useful when the pseudo-inverse is used for the design of an approximate control in the Exact Model Matching.

It gives the definitions of the generalized- and left- inverses of matrices, while it also gives the definitions of the Range- and Null- spaces of matrices, as well.

Generalized- and left- inverses of matrices I

Definition (Generalized-Inverse Matrix)

Given $B_f \in \mathbb{R}^{n\times m}$, matrix $Y_f \in \mathbb{R}^{m\times n}$ that satisfies at least the first or the second of the four equations

$$B_f Y_f B_f = B_f\,, \tag{1a}$$

$$Y_f B_f Y_f = Y_f\,, \tag{1b}$$

$$(B_f Y_f)^\top = B_f Y_f\,, \tag{1c}$$

$$(Y_f B_f)^\top = Y_f B_f\,, \tag{1d}$$

is called the generalized-inverse of B_f (it may be considered the generalization of the inverse of a nonsingular square matrix). $\square$

Any matrix satisfying equations $(a), (b), \ldots,$ from among (1a)-(1d) is called an $\{1, 2, \ldots\}$-inverse of B_f and is denoted by $B_f^{(1,2,\ldots)}$.

Anyhow, every real matrix possesses a unique $\{1, 2, 3, 4\}$-inverse, called the Moore-Penrose inverse or pseudo-inverse, usually denoted by $B_f^\dagger$.

Generalized- and left- inverses of matrices II

Definition (Left-Inverse Matrix)

An $m \times n$ matrix $B_f^\dagger$ such that

$$B_f^\dagger B_f = I_m$$

is called a left-inverse of B_f; obviously, a left-inverse is a $\{1, 2, 4\}$-inverse. $\square$

Range- and Null- spaces of matrices I

Definition (Matrix Image / Range-Space)

Given matrix $A_f \in \mathbb{R}^{n \times n}$, the Image / Range-Space of A_f is defined as the set

$$\mathrm{Im}\,(A_f) \equiv \mathscr{R}(A_f) = \{y_f \in \mathbb{R}^n \mid y_f = A_f x_f, \text{ for some } x_f \in \mathbb{R}^n\}\,.$$

□

Definition (Matrix Kernel / Null-Space)

Given matrix $A_f \in \mathbb{R}^{n \times n}$, the Kernel / Null-Space of A_f is defined as the set

$$\mathrm{Ker}\,(A_f) \equiv \mathscr{N}(A_f) = \{x_f \in \mathbb{R}^n \mid A_f x_f = 0\}\,.$$

□

Classical Pseudo-Inverse Method I

When exact model matching is impossible $\left(\hat{A}_f \neq \hat{A}_n^*\right)$, an approximate solution may be computed by minimizing the criterion

$$J_f(K_f) = \left\| (A_f - B_f K_f) - \hat{A}_n^* \right\|_F^2, \tag{2}$$

where $\|.\|_F$ is the Frobenius-norm of the matrix-argument, i.e., $\|X\|_F = \sqrt{\sum_{i,j} x_{i,j}^2}$.

Classical Pseudo-Inverse Method II

It is simple to show that the control law

$$K_f^* = \arg\min_{K_f} J_f(K_f) \tag{3}$$

is still obtained by using the faulty counterpart of (4), respectively

$$K_n = B_n^\dagger (A_n - \hat{A}_n^*)\,, \tag{4}$$

and that $J_f(K_f^*) = 0$ if and only if the compatibility condition in the faulty case (5), respectively

$$\mathrm{Im}\left(A_f - \hat{A}_n^*\right) \subseteq \mathrm{Im}\,(B_f)\,, \tag{5}$$

holds.

Classical Pseudo-Inverse Method III

The control solution provides to be efficient based on the following classical result on Gerschgorin disks.

Theorem (Varga; 2004)

The eigenvalues of $\hat{A}_f$ lie in the union of the disks (Gerschgorin disks)

$$\mathscr{G}_{f_i} = \left\{ \hat{\lambda}_f \mid |\hat{\lambda}_f - \hat{\lambda}^*_{n_i}| \leq \left\| \hat{Z}^*_n \right\|_2 \left\| \hat{Z}^{*-1}_n \right\|_2 J_f \right\} ,$$

whether $\hat{A}^*_n$ is non-defective, that is it can be reduced to diagonal form by the similarity transformation $\hat{Z}^*_n$, which is its eigenvector matrix, respectively

$$\hat{Z}^{*-1}_n \hat{A}^*_n \hat{Z}^*_n = diag\left(\hat{\lambda}^*_{n_1}, \hat{\lambda}^*_{n_2}, \ldots, \hat{\lambda}^*_{n_n}\right) ,$$

where J_f is defined in (2) and $\|.\|_2$ denotes the 2-norm of the matrix-argument. $\square$

Classical Pseudo-Inverse Method IV

Explicitly, the previous theorem states that the control gain matrix (3), computed using the Pseudo-Inverse Method as in (6), respectively

$$(K_f^{PIM} \equiv)\ K_f = B_f^{\dagger}(A_f - \hat{A}_n^*)\,, \tag{6}$$

guarantees the stability of the post-fault system if the eigenvalues of the nominal closed-loop system are relatively far from the imaginary-axis of the complex-plane and the fault is not too severe such that criterion J_f is small enough.

See also the sufficient condition developed later as in (7), respectively

$$c_{\hat{A}_f} \leq \frac{1}{\|Z_f\|_2 \left\|Z_f^{-1}\right\|_2} \max_{\hat{\lambda}_n^*} \Lambda(\hat{A}_n^*)\,, \text{ for any } \hat{A}_f \in \hat{\mathscr{A}}_f, \tag{7}$$

where $\Lambda(.)$ denotes the spectrum of the matrix-argument.

Classical Pseudo-Inverse Method V

Anyhow, replacing K_f by its optimal value gives

$$(A_f - B_f K_f) - \hat{A}_n^* = \left(I_n - B_f B_f^\dagger\right)\left(A_f - \hat{A}_n^*\right) ,$$

which, when post-fault evaluating (2), leads to the following expression

$$J_f = \sum_{i=1}^{n} \left(a_f^i - \hat{a}_n^{*\,i}\right)^\top \left(I_n - B_f B_f^\dagger\right)\left(a_f^i - \hat{a}_n^{*\,i}\right) , \tag{8}$$

where a_f^i and $\hat{a}_n^{*\,i}$ are the i-th columns of A_f and $\hat{A}_n^*$, respectively; note that J_f is zero when the pair (A_f, B_f) is consistent with (5), but it is nonzero otherwise.

Modified Pseudo-Inverse Method I

The extension of PIM, namely the Modified Pseudo-Inverse Method (MPIM), avoids encountering post-fault closed-loop instability by restricting the computation of K_f such that to guarantee the stability of the post-fault system while achieving as much of the closed-loop nominal performance as possible.

Such an attempt is based on results from the structured perturbation theory, that is it provides an acceptable solution as long as the fault impairs the nominal behavior, but the difference between the two closed-loop matrices $E_f = \hat{A}_f - \hat{A}_n^*$ is still bounded; moreover, it is assumed that matrix E_f may be expressed as $E_f = \sum_{i=1}^{m} c_{f_i} E_{f_i}$, where E_{f_i} are constant matrices and coefficients c_{f_i} vary in intervals around zero.

Modified Pseudo-Inverse Method II

The development for MPIM is based on the following result.

Theorem (Zhou and Khargonekar; 1987)

Let $\hat{X}_f$ *be the solution of the Lyapunov equation*

$$\hat{A}_f^\top \hat{X}_f + \hat{X}_f \hat{A}_f + 2I_n = 0\,, \tag{9}$$

and define $\hat{X}_{f_i}$, $\hat{X}_{f_e}$ *as*

$$\begin{aligned} \hat{X}_{f_i} &= (E_{f_i}^\top \hat{X}_f + \hat{X}_f E_{f_i})/2 = [\hat{X}_f E_{f_i}]_S\,, \;\; \textit{for } i = 1, 2, \ldots, m\,, \\ \hat{X}_{f_e} &= \begin{bmatrix} \hat{X}_{f_1} & \hat{X}_{f_2} & \ldots & \hat{X}_{f_m} \end{bmatrix}\,, \end{aligned}$$

where $[X]_S$ *denotes the symmetric-part of the matrix-argument, i.e.,* $[X]_S = \left(X + X^\top\right)/2$.

Modified Pseudo-Inverse Method III

Theorem (Zhou and Khargonekar; 1987 – cont'd)

Then, $\hat{A}_f$ *is stable if:*

$$\sum_{i=1}^{m} c_{f_i}^2 < \frac{1}{\sigma_{\max}^2(\hat{X}_{f_e})}\,, \quad \textit{or} \tag{10a}$$

$$\sum_{i=1}^{m} |c_{f_i}|\sigma_{\max}(\hat{X}_{f_i}) < 1\,, \quad \textit{or} \tag{10b}$$

$$|c_{f_j}| < \frac{1}{\sigma_{\max}\left(\sum_{i=1}^{m} |\hat{X}_{f_i}|\right)}\,, \quad \textit{for } j = 1, 2, \ldots, m\,, \tag{10c}$$

where $|X|$ *indicates the matrix-argument with its elements in absolute value and* $\sigma_{\max}$ *stands for the maximum singular value, i.e., like* σ_{n_1} *in the SVD theorem.* □

Modified Pseudo-Inverse Method IV

The above result is proved by showing that $V_f[x(t)] = x^\top(t)\hat{X}_f x(t)$ is a Lyapunov function for (11)-(12), respectively

$$M_f : \ \dot{x}(t) = A_f x(t) + B_f u_f(t) \,, \tag{11}$$

and

$$u_f(t) = -K_f x(t) \,, \tag{12}$$

under each of the three conditions in (10); when using the above inequalities as sufficient conditions for the post-fault closed-loop stability and choosing such stability bound δ_f to be, for example, the right hand-side of (10a), condition

$$|k_{f_{i,j}}| \leq \tilde{\delta}_f \,, \ \text{for } i = 1, 2, \ldots, m \,, \ \text{and } j = 1, 2, \ldots, n \,, \tag{13}$$

guarantees the stability of $\hat{A}_f$, where $\tilde{\delta}_f = \delta_f - \epsilon_f$, for some small ϵ_f.

Therefore, assuming that the PIM solution (6) makes the post-fault closed-loop system $\hat{A}_f$ unstable, the problem changes to determining K_f such that to minimize J_f in (2) subject to the stability constraint in (13).

Modified Pseudo-Inverse Method V

Then, when the plant is a single-input system (matrix $B_f \in \mathbb{R}^{n\times 1}$) or it can be decomposed into such single-input subsystems, the following result gives an explicit solution for the post-fault control gain.

Theorem (Gao and Antsaklis; 1991)

Given the pseudo-inverse control matrix K_f^{PIM} as obtained from (6), the modified pseudo-inverse control K_f^{MPIM} solves the reconfigurable control problem with stability guarantees, being computed as

$$k_{f_{i,j}}^{MPIM} = k_{f_{i,j}}^{PIM}, \ \ if \, |k_{f_{i,j}}^{PIM}| \leq \tilde{\delta}_f, \ \ and, \tag{14a}$$

$$k_{f_{i,j}}^{MPIM} = sign\left(k_{f_{i,j}}^{PIM}\right)\tilde{\delta}_f, \ \ otherwise, \tag{14b}$$

where $sign\,(.)$ stands for the sign-function of the real-argument. □

Modified Pseudo-Inverse Method VI

Following from above, as it uses PIM for computing the post-fault control matrix and modifies it under stability constraints to ensure the post-fault closed-loop stability, the previous theorem represents the basis for the so-called Modified Pseudo-Inverse Method.

In any event, though approximate model matching seems reasonable, there are some drawbacks to the standard pseudo-inverse method and its modified pseudo-inverse extension, as their rationale is questionable:

i. The control solution is based on sufficient stability conditions and this may provide very conservative solutions, especially when the number of control inputs is significant. On another hand, it may be difficult to solve the constrained optimization problem in real-time when solutions are urgently needed.

ii. Even when it is stable, finding the post-fault system closest to the nominal closed-loop one does not guarantee that it will exhibit a satisfactory behavior because no performance consideration is included in the problem setting. (The actuators may be overloaded or their mechanical limits may be forced.)

iii. The very meaning of "closest" is questionable, since the choice of the Frobenius norm-based distance is arbitrary. When approximating the nominal closed-loop model, choosing another distance would provide another solution.

Modified Pseudo-Inverse Method VII

Moreover, it follows from the problem setting that any fault (A_f, B_f) can be recovered since there always exists a solution to the minimal-distance problem. This point contradicts the intuitive fact that, in some fault situations, there is no control which can achieve fault recovery.

In this respect, an admissible model matching approach will be proposed in the following.

Outline

1. Pseudo-Inverse vs. Modified Pseudo-Inverse Methods
2. B747 Application Example
   - Pseudo-Inverse Method
   - Modified Pseudo-Inverse Method - Multiplicative Fault
   - Modified Pseudo-Inverse Method - Structural Fault

Pseudo-inverse method I

When the compatibility conditions are no longer satisfied, Pseudo-Inverse Method (PIM) looks for an approximate solution.

The following table displays the pairs (damping coefficient, natural frequency) that are obtained for $\tau_f = 0.9$ and different values of γ_f, whereas it is easily seen that the behavior is very quickly degraded as γ_f increases.

γ_f	0.01	0.02	0.03	0.04	0.05
$k_{f_{m_1}}^{PIM}$	-0.0720	-0.0738	-0.0755	-0.0772	-0.0790
$k_{f_{m_2}}^{PIM}$	-22.0441	-17.7481	-13.4522	-9.1562	-4.8603
$\zeta_{f_m}^{PIM}$	0.6005	0.5149	0.4284	0.3411	0.2528
$\omega_{f_m}^{PIM}$	2.8434	2.8290	2.8146	2.8000	2.7854

γ_f	0.06	0.07	0.08	0.09	0.10
$k_{f_{m_1}}^{PIM}$	-0.0807	-0.0824	-0.0842	-0.0859	-0.0877
$k_{f_{m_2}}^{PIM}$	-0.5643	3.7316	8.0276	12.3236	16.6195
$\zeta_{f_m}^{PIM}$	0.1636	0.0735	-0.0176	-0.1097	-0.2028
$\omega_{f_m}^{PIM}$	2.7707	2.7559	2.7411	2.7261	2.7111

Pseudo-inverse method II

For example, for $\tau_f = 0.9$ and $\gamma_f = 0.1$, the pair (A_f, B_f) is

$$A_f = \begin{bmatrix} -0.2814 & 696.6000 \\ -0.0009 & -0.3856 \end{bmatrix} \text{ and } B_f = \begin{bmatrix} -1.7942 \\ -0.1158 \end{bmatrix}.$$

The PIM approach provides the control-gain matrix

$$K_{f_m}^{PIM} = \begin{bmatrix} -0.0877 & 16.6195 \end{bmatrix},$$

which results in the closed-loop system matrix

$$\hat{A}_{f_m}^{PIM} = \begin{bmatrix} -0.4387 & 726.4181 \\ -0.0110 & 1.5385 \end{bmatrix}.$$

Poles being

$$\Lambda\left(\hat{A}_{f_m}^{PIM}\right) = \{0.5499 \pm 2.6547\jmath\},$$

the PIM-based closed-loop is therefore unstable, with dynamical parameters damping coefficient $\zeta_{f_m}^{PIM} = -0.2028$ and natural frequency $\omega_{f_m}^{PIM} = 2.7111$ rad/sec.

Modified pseudo-inverse method - Multiplicative fault I

Whether encountering instability using PIM, addressing the stability problem is the goal of the Modified Pseudo-Inverse Method (MPIM).

The best stable closed-loop matrix obtained with MPIM is

$$\hat{A}_{f_m}^{MPIM} = \begin{bmatrix} -0.3415 & 696.6600 \\ -0.0048 & -0.3817 \end{bmatrix},$$

which is associated with the feedback control matrix

$$K_{f_m}^{MPIM} = \begin{bmatrix} -0.0335 & 0.0335 \end{bmatrix},$$

Modified pseudo-inverse method - Multiplicative fault II

obtained following the MPIM algorithm from (14) with $\tilde{\delta}_{f_m} = 0.0335$ from the SVD of $B_f K_{f_m}^{PIM}$, precisely $U_{f_m} \Sigma_{f_m} V_{f_m}^{\top}$ as

$$U_{f_m} = \begin{bmatrix} -0.9979 & -0.0644 \\ -0.0644 & 0.9979 \end{bmatrix},$$

$$\Sigma_{f_m} = \begin{bmatrix} 29.8805 & 0 \\ 0 & 0.0000 \end{bmatrix},$$

$$V_{f_m} = \begin{bmatrix} -0.0053 & 1.0000 \\ 1.0000 & 0.0053 \end{bmatrix},$$

with $\sigma_{f_m}^{\max} = 29.8805$.

Note that this approach indeed allows to stabilize the system but offers no real control of the behavior of the closed-loop system which results from fault accommodation, as the closed-loop spectrum

$$\Lambda\left(\hat{A}_{f_m}^{MPIM}\right) = \{-0.3616 \pm 1.8237\jmath\},$$

leads to a damping coefficient $\zeta_{f_m}^{MPIM}$ of about 0.2 and a natural frequency $\omega_{f_m}^{MPIM}$ of approximately 1.85 rad/sec, performance that is not acceptable.

Modified pseudo-inverse method - Multiplicative fault III

The following table shows the evolution of the damping coefficient and natural frequency for $\gamma_f = 0.1$ under increasing values of τ_f.

τ_f	0.1	0.2	0.3	0.4
$k_{f_{m_1}}^{MPIM}$	-0.0097	-0.0110	-0.0125	-0.0146
$k_{f_{m_2}}^{MPIM}$	0.0335	0.0335	0.0335	0.0335
$\zeta_{f_m}^{MPIM}$	0.1408	0.1415	0.1422	0.1429
$\omega_{f_m}^{MPIM}$	2.8027	2.8029	2.8031	2.8033

τ_f	0.5	0.6	0.7	0.8	0.9
$k_{f_{m_1}}^{MPIM}$	-0.0175	-0.0219	-0.0292	-0.0335	-0.0335
$k_{f_{m_2}}^{MPIM}$	0.0335	0.0335	0.0335	0.0335	0.0335
$\zeta_{m_f}^{MPIM}$	0.1435	0.1442	0.1449	0.1568	0.1945
$\omega_{m_f}^{MPIM}$	2.8035	2.8037	2.8039	2.4855	1.8592

Modified pseudo-inverse method - Structural fault I

Considering again the system under actuator faults with $\tau_f = 0.5$ but also under structural impairments as $A_f = A_n + \Delta A_f \times I_n$, with $\Delta A_f = \sigma_n^{\max} \times 10^{-3}$, where $\sigma_n^{\max} = 774.0000$ is the maximum singular value of A_n, it can then be seen that MPIM provides the closed-loop control

$$K_{f_s}^{MPIM} = \begin{bmatrix} -0.0210 & -0.0210 \end{bmatrix},$$

from

$$K_{f_s}^{PIM} = \begin{bmatrix} -0.0999 & -5.2735 \end{bmatrix},$$

Modified pseudo-inverse method - Structural fault II

and the SVD of $B_f K_{f_s}^{PIM}$ as $U_{f_s} \Sigma_{f_s} V_{f_s}^{\top}$, precisely

$$U_{f_s} = \begin{bmatrix} -0.9979 & -0.0644 \\ -0.0644 & 0.9979 \end{bmatrix},$$

$$\Sigma_{f_s} = \begin{bmatrix} 47.4148 & 0 \\ 0 & 0.0000 \end{bmatrix},$$

$$V_{f_s} = \begin{bmatrix} -0.0189 & 0.9998 \\ -0.9998 & -0.0189 \end{bmatrix},$$

with $\sigma_{f_s}^{\max} = 47.4148$, which results in the closed-loop system matrix

$$\hat{A}_{f_s}^{MPIM} = \begin{bmatrix} 0.2730 & 773.8117 \\ -0.0132 & 0.3334 \end{bmatrix}.$$

Poles being

$$\Lambda\left(\hat{A}_{f_s}^{MPIM}\right) = \{0.3032 \pm 3.1898\jmath\},$$

the MPIM-based closed-loop is therefore unstable, with dynamical parameters damping coefficient $\zeta_{f_s}^{MPIM} = -0.0946$ and natural frequency $\omega_{f_s}^{MPIM} = 3.20$ *rad/sec*.

Modified pseudo-inverse method - Structural fault III

It can also be seen in the following table through the dynamical parameters, namely the damping coefficient and natural frequency, that, for the structural fault introduced and under increasing values of τ_f, the post-fault system remains unstable even under $K_{f_s}^{MPIM}$.

τ_f	0.1	0.2	0.3	0.4
$\zeta_{f_s}^{MPIM}$	-0.0375	-0.0457	-0.0551	-0.0661
$\omega_{f_s}^{MPIM}$	4.9945	4.7185	4.4253	4.1113

τ_f	0.5	0.6	0.7	0.8	0.9
$\zeta_{f_s}^{MPIM}$	-0.0795	-0.0966	-0.1196	-0.1546	-0.2205
$\omega_{f_s}^{MPIM}$	3.7712	3.3973	2.9767	2.4859	1.8706

Regarding this situation, it is then obvious that the stabilizing gain provided by any linear quadratic regulator technique must first be found using an efficient computational technique and just afterwards to adjust it using the modified pseudo-inverse.

Questions? vs. Answers!

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